

Set: Any collection of well-defined objects is called a set. The objects in the set are called Elements.

Null Set: A Set with no elements is said to be null set and it is denoted by ϕ .

$$\text{Null Set : } \phi = \{ \}$$

Power Set: Let A be any set, the set of all subsets of A is said to be power set of A . It is denoted by $P(A)$ (or) $\mathcal{P}(A)$.

$$A = \{1, 2\}$$

$$P(A) \text{ (or) } \mathcal{P}(A) = \{ \{ \}, \{1\}, \{2\}, \{1, 2\} \}$$

* If any set contains n elements then

$$P(A) = 2^n \text{ elements.}$$

Universal Set: A Set is said to be universal set if it contains every element under discussion and it is denoted by ' U ' or ' μ '.

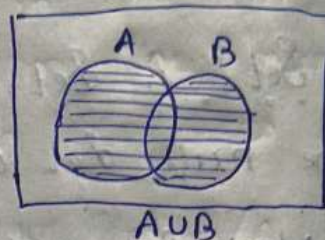
* If A is any non-empty set then

$$A^c = \mu - A$$

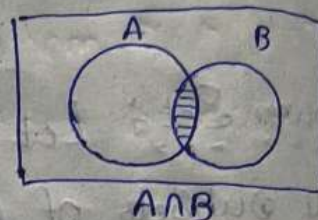
Venn Diagram: The pictorial representation of a Set is called venn diagram.

Operations on Set:

1) Union: If $x \in A$ (or) $x \in B$
then $x \in A \cup B$

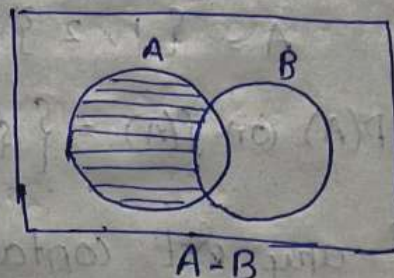


2) Intersection: If $x \in A$ or $x \in B$
then $x \in A \cap B$



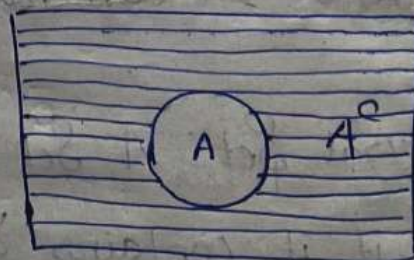
3) Difference:

If $x \in A$ or $x \notin B$
then $x \in (A - B)$



4) Complement:

If $x \notin A$ & $x \in \mu$
then $x \in A^c$

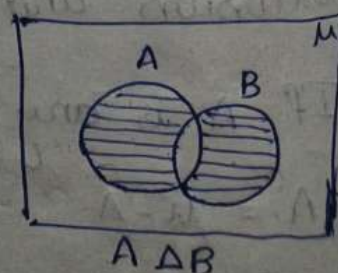


5) Symmetric Difference:

$$A \Delta B = (A \cup B) - (A \cap B)$$

(or)

$$= (A - B) \cup (B - A)$$



Note:

1) $A -$

2) $A -$

3) $(A$

Principle

If

$n(A \cup B)$

* $n(A \cup B)$

* $n(A \cup B)$

In a

football

found

Find

Sol: by

Note:

$$1) A - (B \cup C) = (A - B) \cap (A - C)$$

$$2) A - (B \cap C) = (A - B) \cup (A - C)$$

$$3) (A \cup B)^c = A^c \cap B^c$$

Principle of Inclusion and Exclusion:

If A and B be any two sets then

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$* n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

$$* n(A_1 \cup A_2 \cup \dots \cup A_k) = \sum_{i=1}^k n(A_i) - \sum_{i,j=1}^k n(A_i \cap A_j) + \dots$$

In a class of 50 students, 20 students play football, 16 students play Hockey. If it is found that 10 students play f.B and Hockey. Find the no. of students who played neither.

Sol: Given, $n(A) = 20$ & $n(B) = 16$

$$n(A \cap B) = 10$$

$$n(\text{play both games}) = n(A \cup B) = n(A) + n(B) - n(A \cap B) \\ = 26$$

$$n(\text{Play neither games}) = n(A \cup B) = \mu - n(A \cap B)$$

$$= 50 - 26 = 24$$

Q) Among 60 Students in a class, 45 passed in 1st Sem, 30 passed in 2nd Sem, 12 did not pass in either Sem. How many Students passed in both Semesters.

Sol: Given, no. of passed in 1st Sem $n(A) = 45$

no. of Students passed in 2nd Sem $n(B) = 30$

no. of Students passed in neither Semesters

$$n(A \cup B)^c = \mu - n(A \cup B)$$

no. of Students passed in both Semesters $= n(A \cap B)$

$$n(A \cup B) = \mu - n(A \cap B)$$

$$= 60 - 12 = 48$$

Q) Sol: no. of integers divisible by 7 not exceeding 2000 is $n(A) = 285$

$$n(B) = 181$$

no. of integers divisible by both 7 & 11 =

$$= n(A \cap B)$$

$$= 23$$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B) = 285 + 181 - 25 = 441$$

2) Determine the no. of primes not exceeding 100 and ~~not~~ divisible by 2, 3, 5 or 7

Sol: no. of primes divisible by 2 and not exceeding 100 = $n(p) = n\left(\frac{100}{2}\right) = 50$

no. of primes divisible by 3 and not

Exceeding 100 = $n(p) = n\left(\frac{100}{3}\right) = 33$

no. of " " by 5 " = $n(c) = n\left(\frac{100}{5}\right) = 20$

no. of " " by 7 " = $n(d) = n\left(\frac{100}{7}\right) = 14$

Cartesian Product: Cartesian product of two sets A and B is denoted by $A \times B$ (read as A cross B) and is given by $A \times B = \{(x, y) \mid x \in A \text{ \& } y \in B\}$.

Ex: $A = \{1, 2\}$; $B = \{3, 4\}$

$$A \times B = \{(1, 3), (1, 4), \dots\}$$

Note: If $n(A) = m$, $n(B) = n$

$$n(A \times B) = mn$$

* If A is any non-empty set then $\mu = A \times A$.

Relation: A set of ordered pairs is called a relation or binary operation.

→ If x is related to y then we write it as $\Rightarrow xRy$ or $(x,y) \in R$.

Properties of Relations:

Relations having basic properties and they are

- 1) Reflexive property
- 2) Symmetric property
- 3) Anti-Symmetric property
- 4) Transitive
- 5) Irreflexive

1) Reflexive Property: A relation R is said to be reflexive if $\forall x \in A \Rightarrow (x,x) \in R$.

Ex: ' $\leq$ ', ' $\subseteq$ '

2) Symmetric Property: A relation R is said to be symmetric if $\forall (x,y) \in R \Rightarrow (y,x) \in R$.

3) Anti-Symmetric Property: A relation R is said to be " if $\forall (x, y) \in R \ \& \ (y, x) \in R \Rightarrow x = y$

4) Transitive Property: A relation R is said to be " if $\forall (x, y) \in R \ \& \ (y, z) \in R \Rightarrow (x, z) \in R$

5) Irreflexive Property: A relation R is said to be " if $\forall x \in A \Rightarrow (x, x) \notin R$

Representation of a Relation:

A Relation can be represented by either matrix (a) diagram.

In matrix form:

- * If there is a pair with same Element or a relation with another element take it as 1.
- * If there is no relation b/n two elements then take it as 0.

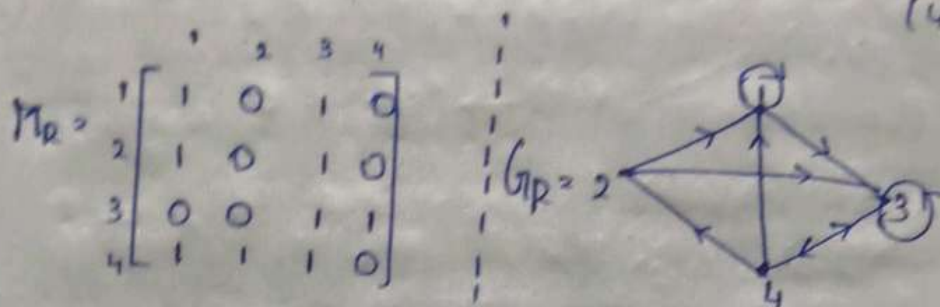
In Graphical form:

- * If there is a relation with same element then draw a loop with that element.

* If there is relation b/n two elements, draw a line b/n that elements.

Eg:

$$R = \{(1,1), (1,3), (2,1), (2,3), (3,3), (3,4), (4,1), (4,2), (4,3)\}$$



Q) If $R = \{(1,1), (1,2), (2,2), (2,3), (3,2)\}$
 $S = \{(1,2), (3,2), (2,3), (3,1)\}$

Sol: i) $R - S = \{(1,1), (2,2)\}$

ii) $S - R = \{(3,1)\}$

domain of $R - S = \{1, 2\}$ [$\because$ elems of x]

Co-domain of $R - S = \{1, 2\}$ [$\because$ elems of y]

Q) Let $A = \{1, 2, 3\}$ &

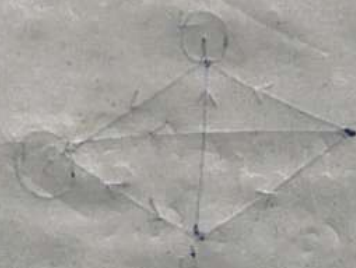
$$R = \{(1,1), (1,3), (2,2), (2,3), (3,1), (3,3)\}$$

$$S = \{(1,3), (2,1), (2,3), (3,2)\}$$

Then find $R \circ S$ & R^2 .

8a) $ROS = \{(1,3), (1,2), (2,1), (2,3), (2,2), (3,3), (3,2)\}$

$$R^2 = R \circ R =$$



Types of Relations:

- 1) Compatible relation: A relation R is said to be compatible relation if it is said to be reflexive and symmetric.
- 2) Equivalence Relation: A relation R is said to be Equivalence relation if it is reflexive, symmetric and transitive.

(3,2)}
3) Partial Ordered Relation: A Relation R is said to be POR if it is reflexive, Anti-Symmetric and transitive.

Q) Let $A = \{1, 2, 3, 4\}$

$$R = \{(1,1), (1,2), (2,1), (2,2), (3,1), (3,3), (1,3), (4,1), (4,4)\}$$

is R is an Equivalence relation.

Sol: Reflexive:

Here $\forall x \in A$ then $(x,x) \in R$

Symmetric
Transitive:

Here $(4,1) \in R$ but $(1,4) \notin R$

$\therefore R$ is not Symmetric

Hence, R is not an Equivalence Relation.

Q) $A = \{1, 2, 3, 4\}$

$$R = \{(1,1), (1,2), (2,1), (1,4), (2,2), (3,3), (4,1), (4,4)\}$$

Check for R is Equivalence Relation.

Sol: Reflexive:

Here $\forall x \in A$ then $(x,x) \in R$

$\therefore R$ is an reflexive.

Symmetric: Here $\forall (x, y) \in R$ then $(y, x) \in R$

$\therefore R$ is Symmetric relation.

Transitive:

Here $(1, 1) \begin{cases} (1, 2) \rightarrow (1, 2) \in R \\ (1, 4) \rightarrow (1, 4) \in R \end{cases}$

$(1, 2) \begin{cases} (2, 1) \rightarrow (1, 1) \in R \\ (2, 2) \rightarrow (1, 2) \in R \end{cases}$

$(2, 1) \begin{cases} (1, 1) \rightarrow (2, 1) \in R \\ (1, 2) \rightarrow (2, 2) \in R \\ (1, 4) \rightarrow (2, 4) \notin R \end{cases}$

$\therefore R$ is not transitive Relation.

Hence, R is not an Equivalence relation but R is compatible relation.

a) Prove that Relation $R = \{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3), (4, 4)\}$. Define on a Set $A = \{1, 2, 3, 4\}$ is an Equivalence relation.

Sol:

Reflexive: Here $\forall x \in A$ then $(x, x) \in R$

$\therefore R$ is an reflexive.

Symmetric: Here $\forall (x, y) \in R$ then $(y, x) \in R$

$\therefore R$ is Symmetric.

Transitive:

$$(2,2) \rightarrow (2,3) \rightarrow (2,3) \in R$$

Q) If $A = \{1,2,3\}$ and $R = \{(1,1), (1,2), (2,2), (2,3), (1,3), (3,3)\}$ is R an Equivalence relation or partial ordered relation.

Sol: Reflexive:

Here $\forall x \in A$ then $(x,x) \in R$

$\therefore R$ is a reflexive.

Symmetric:

Here $(1,2) \in R$ but $(2,1) \notin R$

$\therefore R$ is not Symmetric.

Transitive:

Here $(1,1) \begin{cases} (1,2) \rightarrow (1,2) \in R \\ (1,3) \rightarrow (1,3) \in R \end{cases}$

$(1,2) \begin{cases} (2,2) \rightarrow (1,2) \in R \\ (2,3) \rightarrow (1,3) \in R \end{cases}$

$$(2,2) \rightarrow (2,3) \rightarrow (2,3) \in R$$

$$(2,3) \rightarrow (3,3) \rightarrow (2,3) \in R$$

$$(1,3) \rightarrow (3,3) \rightarrow (1,3) \in R$$

$\therefore R$ is a transitive.

Anti-Symmetric:

Here $\forall (x,y) \in R$ & $(y,x) \in R$ then $x=y$

$\therefore R$ is an Anti-Symmetric.

Hence, R is a Partial Ordered relation.

Q) $x = \{1, 2, 3, 4, 5, 6, 7\}$ and $R = \{(x,y) \mid (x-y) \text{ is divisible by } 3\}$. Show that R is an Equivalence Relation.

Sol: Reflexive:

Here $\forall x \in A$ then $(x,x) \in R$ [$\because (x-x)$ is $\div 3$]

Symmetric:

Here $\forall (x,y) \in R$ i.e., $(x-y)$ is $\div 3$

$$\Rightarrow -(y-x) \text{ is } \div 3$$

$$\Rightarrow (y,x) \in R$$

Transitive:

Here $\forall (x,y) \in R$ & $(y,z) \in R$

$$\text{i.e., } (x-y) \text{ is } \div 3 \rightarrow \textcircled{1}$$

$$(y-z) \text{ is } \div 3 \rightarrow \textcircled{2}$$

$$\text{Now } (x, z) = x - z$$

$$= x - y + y - z$$

$$= (x - y) + (y - z) \text{ is } \div 3$$

$$[\because \textcircled{1} + \textcircled{2}]$$

$\therefore R$ is Transitive.

Hence, R is Equivalence relⁿ.

Q) Let R be a relation on the Set of all ordered pairs of +ve integers, Let it be defined as $(x, y) R (u, v) \Leftrightarrow xv = yu$. Show that R is an Equivalence relation.

Sol: Reflexive:

$$\forall (x, x) R (x, x) \Rightarrow x^2 = x^2 \text{ which is true}$$

$\therefore R$ is Reflexive.

Symmetric:

$$\forall (x, y) R (u, v) \Rightarrow xv = yu$$

$$yu = xv$$

$$uy = vx$$

$$\Rightarrow (u, v) R (x, y)$$

$\therefore R$ is Symmetric.

Transitive:

$$\text{Let } (x, y) R (u, v) \Leftrightarrow xv = yu \rightarrow \textcircled{1}$$

$$(u, v) R (l, m) \Leftrightarrow um = vl \rightarrow \textcircled{2}$$

$$\textcircled{1} \times \textcircled{2} \Rightarrow x \vee . u m, y u . v l$$

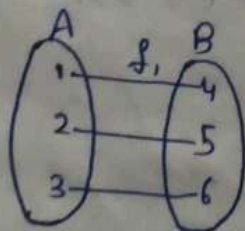
$$x m = y l \Leftrightarrow (x, y) \in R(l, m)$$

Function: Let A and B be two sets, a function f from A to B ($f: A \rightarrow B$) is an assignment of exactly one element of B to each element of A . Here A is called domain and B is called co-domain.

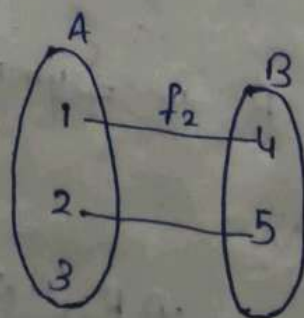
One-One function: Suppose $f: A \rightarrow B$ is a function, it is said to be 1-1 function if $f(a) = f(b) \Rightarrow a = b$ and this is also called injective mapping.

Onto function: Suppose $f: A \rightarrow B$ be a given function, we say that f is onto if for every $b \in B \exists a \in A \Rightarrow f(a) = b$.

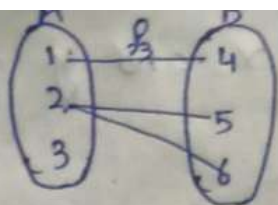
Ex:



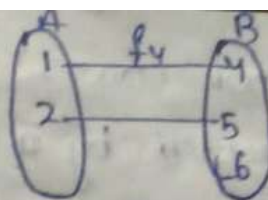
Here f_1 is 1-1



Here f_2 is not 1-1 because '3' has no image



f_3 is onto



Here f_4 is not onto
because '6' has no image.

Bijjective function: A function f is said to be bijective function if f is 1-1 and onto.

POSET (Partial ordered Set): A Set together with partial ordering relation (R) is called POSET.

Ex: $(\mathbb{Z}, \leq)$, $(P(A), \subseteq)$

Hasse Diagram: The pictorial representation of a finite POSET is called Hasse Diagram.

Working Procedure to draw Hasse Diagram:

- 1) Each element is represented by a small circle.
- 2) The circle $x \in P$ is drawn below the $y \in P$ if $x < y$.
- 3) A line is drawn b/w x & y if 'y' covers 'x' and if $x < y$ but

'y' does not covers 'x' then x and y are not connected by a line.

Q) Draw the Hasse diagram for $R = \{(a,b) \mid a \text{ divides } b\}$ where $a = \{1, 2, 3, 4, 5, 6, 8\}$

Sol:

$$R = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (1,8), (2,4), (2,6), (2,8), (3,6), (4,8), (2,2), (3,3), (4,4), (5,5), (6,6), (8,8)\}$$

Reflexive:

$\forall x \in A$ then $(x,x) \in R$
 $\therefore R$ is a Reflexive

Anti-Symmetric:

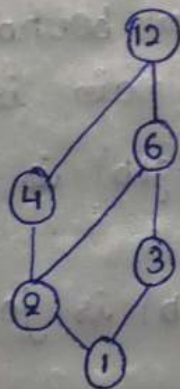
Here $\forall (x,y) \in R$ & $(y,x) \in R$
 $\therefore x = y$

Sol:

Q) Draw the Hasse diagram of $(D_{12}, |)$

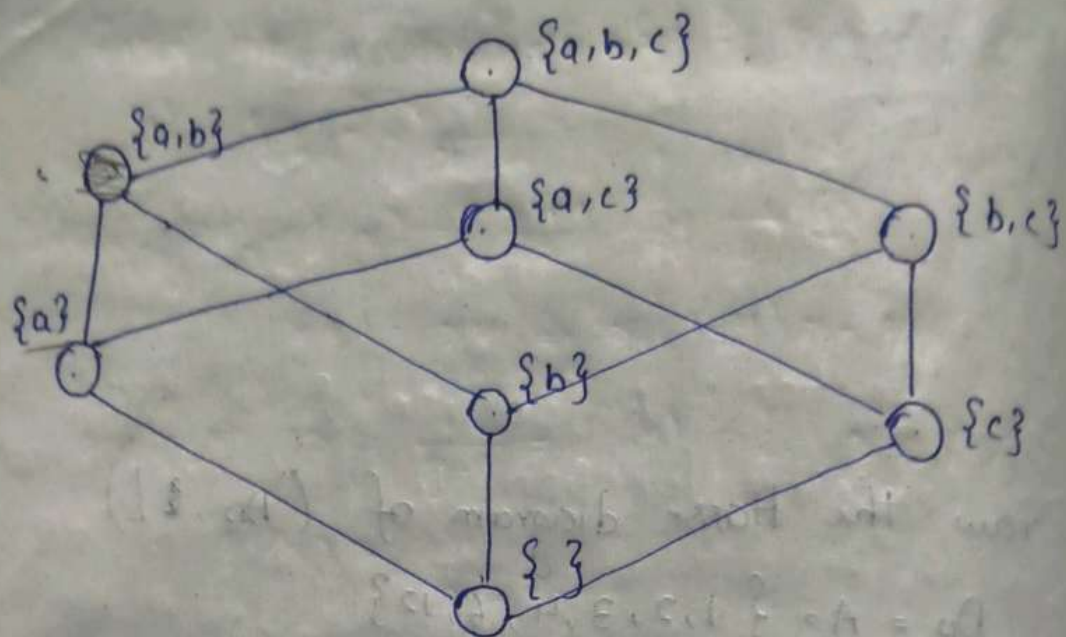
Sol: $D_{12} = A = \{1, 2, 3, 4, 6, 12\}$

Hasse Diagram:



Q) Draw the Hasse diagram of POSET $\{P(S), \subseteq\}$ where $S = \{a, b, c\}$.

Sol: $P(S) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$



Lattice: A POSET of n elements ($n > 1$) which has both greatest lower bound and least upper bound for each pair is called Lattice.

- * Greatest lower bound (glb) is meet or product ($\wedge$ or $*$).
- * Least upper bound (lub) is join or sum ($\vee$ or $+$).

Types of Lattice:

- 1) Distributive Lattice
- 2) Modular Lattice

1) Distributive Lattice (DL): A Lattice is said to be DL if it satisfies the following distributive properties.

- i) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$
- ii) $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

2) Modular Lattice: A Lattice is said to be ML if it satisfies the following prop.
 $a \leq c \Rightarrow a \vee (b \wedge c) = (a \vee b) \wedge c$

Q) Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be a function defined as $f(x) = 3x + 5$. Check f is a bijective func.

Sol: Now $f(x_1) = f(x_2) \Rightarrow 3x_1 + 5 = 3x_2 + 5$
 $\Rightarrow 3x_1 = 3x_2$
 $\Rightarrow x_1 = x_2$

$\therefore f$ is 1-1 function.

Here $f(x) = 3x + 5 = y$

$3x = y - 5$

$x = \frac{y-5}{3} \notin \mathbb{Z}$

$\therefore f$ is not onto func.

$\therefore f$ is not a bijective function.

Q) Let the function $f: \mathbb{N} \rightarrow \mathbb{N}$ defined as $f(n) = n + 2$ determine the function f is bijective or not.

Sol: Now, $f(n_1) = f(n_2) \Rightarrow n_1 + 2 = n_2 + 2$
 $n_1 = n_2$

$\therefore f$ is one-one

Here, $f(n) = n + 2 = m$

$n = m - 2 \notin \mathbb{N}$

f is not a bijective.

Q) Find the inverse of the following functions

1) $f(x) = \frac{x+1}{x}$

Sol: Given, $f(x) = \frac{x+1}{x} = y \rightarrow \text{①}$

$$x+1 = xy \Rightarrow x(1-y) = -1$$

$$\Rightarrow x = \frac{1}{y-1} = f^{-1}(y) \quad [\because x = f^{-1}(y)]$$

$$\therefore f^{-1}(x) = \frac{1}{x-1} //$$

2) $f(x) = x^3 + 2 = y \rightarrow \text{①}$

$$x^3 = y - 2$$

$$\Rightarrow x = (y-2)^{1/3} = f^{-1}(y)$$

$$f^{-1}(y) = (x-2)^{1/3} //$$

3) $f(x) = 4e^{6x+2} = y$

$$e^{6x+2} = \frac{y}{4} \Rightarrow 6x+2 = \log\left(\frac{y}{4}\right)$$

$$x = \frac{1}{6} \log\left(\frac{y}{4}\right) - \frac{1}{3} = f^{-1}(y)$$

Composition of functions: Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be two fun's then the composition f and g is denoted by $\underline{fog}$ and is the func from $A \rightarrow C$ & is defined as $(fog)x = f[g(x)] \quad \forall x \in A$.

Sol:

Different Types of Composition Functions:

i) $(g \circ f)(x) = g[f(x)]$

ii) $(f \circ g) \circ h(x) = f[g[h(x)]]$

iii) $f^2(x) = f[f(x)]$

Q) Let f, g be two funcs from $\mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = 2x$ & $g(x) = 3x^2 - 5$ then find $f \circ g, g \circ f, f^2$.

Sol: $(f \circ g)(x) = f(g(x))$
 $= f(3x^2 - 5)$
 $= 2(3x^2 - 5) = 6x^2 - 10$

$$(g \circ f)(x) = g(f(x))$$
$$= g(2x) = 3(2x)^2 - 5 = 12x^2 - 5$$

$$f^2(x) = f(f(x))$$
$$= 2(2x) = 4x$$

Q) Let $x = \{1, 2, 3\}$; f, g be two funcs from $x \rightarrow x$ given by $f = \{(1, 2), (2, 3), (3, 1)\}$ & $g = \{(1, 2), (2, 1), (3, 3)\}$ then find $f \circ g, g \circ f, f^2$.

Sol: $f \circ g(1) = f[g(1)]$
 $= f[2] = 3$

$$f \circ g(2) = f[g(2)]$$
$$= f[1] = 2$$

$$f \circ g(3) = f[g(3)] \\ = f[3] = 1$$

$$2) g \circ f(1) = g[f(1)] \\ = g[2] = 1$$

$$g \circ f(2) = g[f(2)] \\ = g[3] = 3$$

$$g \circ f(3) = g[f(3)] \\ = g[1] = 2$$

$$3) f^2(1) = f[f(1)] \\ = f[2] = 3$$

$$f^2(2) = f[f(2)] \\ = f[3] = 1$$

$$f^2(3) = f[f(3)] \\ = f[1] = 2$$

$$\therefore f \circ g = \{(1, 3), (2, 2), (3, 1)\}$$

$$g \circ f = \{(1, 1), (2, 3), (3, 2)\}$$

$$f \circ f = \{(1, 3), (2, 1), (3, 2)\}$$

Permutation function:

Set of all bijective functions $\rho: A \rightarrow A$ is known as 'Permutation functions'.

$$\text{Eg: } A = \{a, b, c\}$$

$$P_1 = \begin{pmatrix} a & b & c \\ a & b & c \end{pmatrix}$$

$$P_3 = \begin{pmatrix} a & b & c \\ b & c & a \end{pmatrix}$$

$$P_2 = \begin{pmatrix} a & b & c \\ a & c & b \end{pmatrix}$$

$$P_4 = \begin{pmatrix} a & b & c \\ b & a & c \end{pmatrix}$$

$$P_3 = \begin{pmatrix} a & b & c \\ c & a & b \end{pmatrix}$$

$$P_6 = \begin{pmatrix} a & b & c \\ c & b & a \end{pmatrix}$$

Note:

If a set contains n elements then the total no. of permutations = $n!$.

Q) Let $A = \{1, 2, 3, 4\}$ and $f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}$ & $g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}$. Then find $f \circ g$, $g \circ f$, f^2 .

Sol:

$$f \circ g(1) = f[g(1)] = f(4) = 3$$

$$f \circ g(2) = f[g(2)] = f(1) = 2$$

$$f \circ g(3) = f[g(3)] = f(2) = 4$$

$$f \circ g(4) = f[g(4)] = f(3) = 1$$

$$\therefore f \circ g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix}$$

$$g \circ f(1) = 1 ; g \circ f(2) = 3 ; g \circ f(3) = 4 ; g \circ f(4) = 2$$

$$\therefore g \circ f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 4 & 2 \end{pmatrix}$$

$$f^2(1) = 4 ; f^2(2) = 3 ; f^2(3) = 2 ; f^2(4) = 1$$

$$\therefore f^2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$$

Inverse Permutation: If there is a permutation f on $A \rightarrow f^{-1}$ on A such that $f \circ f^{-1} = f^{-1} \circ f = I$. Here f^{-1} is called inverse permutation of f . Where I is the Identity permutation.

Eg: $f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} \Rightarrow f^{-1} = \begin{pmatrix} 4 & 3 & 2 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix}$

Transitive closure: Transitive closure of a binary relation R on a set S is the smallest transitive relation on ' S ' which contains ' R ' and is denoted by R^+ and is evaluated by $R^+ = R \cup R^2 \cup R^3 \cup \dots \cup R^n$ where $R^n = \{ \}$.

Q) Find the TC of the relation $R = \{(a,b), (b,c), (c,d), (d,e)\}$ defined on set $A = \{a, b, c, d, e\}$

Sol: $R \cup R^2 = \{(a,b), (b,c), (c,d), (d,e)\} \cup \{(a,c), (b,d), (c,e)\}$

$R^2 = \{(a,c), (b,d), (c,e)\}$

$R^3 = R^2 \cup R = \{(a,c), (b,d), (c,e)\} \cup \{(a,b), (b,c), (c,d), (d,e)\}$

$R^3 = \{(a,d), (b,e)\}$

$$R^4 = R^3 \circ R = \{(a, e)\}$$

$$R^5 = \{\}$$

$$\therefore \text{Transitive closure } R^+ = R \cup R^2 \cup R^3 \cup R^4 \cup R^5 \\ = \{(a, b), (b, c), (c, d), (d, e), (a, c), (b, d), (c, e), \\ (a, d), (b, e), (a, e)\}$$

Properties of Lattice:

* Idempotent Law: i) $a \vee a = a$
ii) $a \wedge a = a$

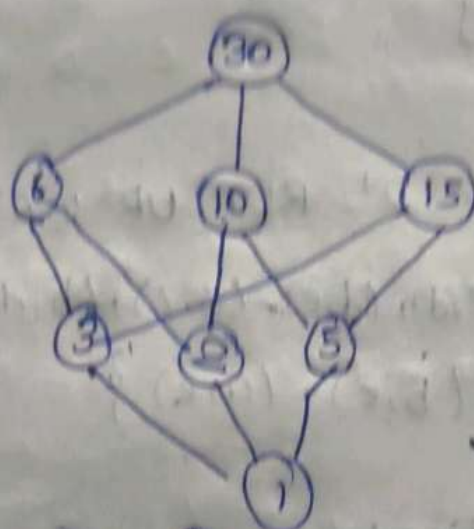
* Commutative Law: i) $a \vee b = b \vee a$
ii) $a \wedge b = b \wedge a$

* Associative Law: i) $a \vee (b \vee c) = (a \vee b) \vee c$
ii) $a \wedge (b \wedge c) = (a \wedge b) \wedge c$

* Absorption Law: i) $a \vee (a \wedge b) = a$
ii) $a \wedge (a \vee b) = a$

Q) Draw the Hasse diagram of $(D_{30}, 1)$.

sol: $D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$

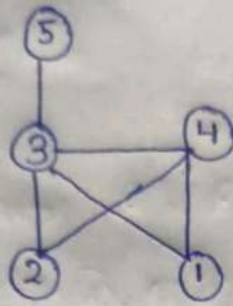


2) Draw the Hasse diagram for the relation defined on $A = \{1, 2, 3, 4, 5\}$ whose relation

matrix is given $M_R =$

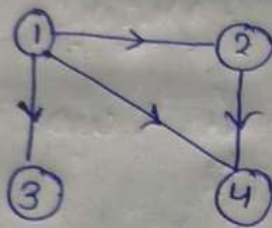
	1	2	3	4	5
1	1	0	1	1	1
2	0	1	1	1	1
3	0	0	1	1	1
4	0	0	0	1	0
5	0	0	0	0	1

Sol: Here, $R = \{(1,1), (1,3), (1,4), (1,5), (2,2), (2,3), (2,4), (2,5), (3,3), (3,4), (3,5), (4,4), (5,5)\}$



Hasse Diagram

a) Draw the Hasse Diagram for the following graph



Sol: Here $R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4)\}$

